

Exercise #7

27. February 2023

Problem 1. (PDEs)

- a) Check that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x)$ solves the following equation on $[0, 1]$,

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x),$$

where $u(0) = u(1) = 0$.

- b) Determine the constant $\alpha \in \mathbb{R}$ so that $u = \frac{1}{16\pi^4 - 4\pi^2} \sin(2\pi x) + \alpha \sin(4\pi x)$ solves,

$$\partial_x^4 u + \partial_x^2 u = \sin(2\pi x) + \sin(4\pi x), \quad u(0) = u(1) = 0.$$

Solution.

- a) We insert u into the equation and check that it holds: We have,

$$\partial_x^2 u = -4\pi^2 u, \quad \partial_x^4 u = 16\pi^4 u.$$

Inserted into the equation, we see that $(\partial_x^4 + \partial_x^2)u = (16\pi^4 - 4\pi^2)u = \sin(2\pi x)$, which was what we wanted to show.

- b) By superposition and the last problem, we know that the first term in u gives the first term in the right hand side in the equation. Therefore, we can focus adapting α to the second term. We have,

$$(\partial_x^4 + \partial_x^2) \sin(4\pi x) = (16^2\pi^4 - 16\pi^2) \sin(4\pi x).$$

Therefore, if we choose $\alpha = (16^2\pi^4 - 16\pi^2)^{-1}$ we should get the desired result.

Problem 2. (Superposition)

In this problem we will consider the heat equation on $[0, 1]$:

$$\partial_t u - \partial_x^2 u = f(x). \quad (1)$$

- a) Consider (1) when $f(x) = 0$. Show that,

$$u_1 = e^{-4\pi^2 t} \cos(2\pi x), \quad \text{and} \quad u_2 = e^{-16\pi^2 t} \cos(4\pi x),$$

solves (1) in this case.

- b) Argue that $u_1 + u_2$ is also a solution to (1) in this case.
 c) Set $f(x) = \sin(2\pi x)$. Show that,

$$u_3 = \frac{1}{4\pi^2} (1 - e^{-4\pi^2 t}) \sin(2\pi x),$$

is a solution to (1) in this case. Is $u_3 + u_3$ a solution to (1) in this case?

- d) We now impose boundary conditions and an initial condition on (1):

$$u(0, x) = g(x), \quad u(t, 0) = u(t, 1) = 0.$$

Set $f(x) = \sin(2\pi x)$ and $g(x) = 0$, and check that u_3 (from point c) solves (1) with these boundary conditions, and the initial condition.

- e) Now, set $f(x) = 0$ and $g(x) = \sin(4\pi x)$, and denote the solution to (1) by u_4 . Check that,

$$u_4 = e^{-16\pi^2 t} \sin(4\pi x).$$

Solution.

- a) We compute $(\partial_t - \partial_x^2)u_1 = (-4\pi^2 + 4\pi^2)u_1 = 0$, and $(\partial_t - \partial_x^2)u_2 = (-16\pi^2 + 16\pi^2)u_2 = 0$.
- b) Since (1) is linear and homogeneous, we can use superposition. Therefore, $u_1 + u_2$ is a solution to (1).
- c) We compute $\partial_t u_3 = -4\pi^2 e^{-4\pi^2 t} \sin(2\pi x)$. On the other hand, we have $\partial_x^2 u_3 = -(1 - e^{-4\pi^2 t}) \sin(2\pi x)$. Subtracting these two, gives $\sin(2\pi x)$.

$u_3 + u_3$ would not be a solution to (1) in this case, since the equation is no longer homogeneous. (It would on the other hand be a solution to (1) with $f(x) = 2 \sin(2\pi x)$).

- d) In order to check if $u_3(t, x)$ satisfies (1), we first check the boundary conditions. We have $u_3(t, 0) = u_3(t, 1) = 0$, since $\sin(x) = 0$ at $x = 0, 2\pi$.

We then check the initial condition: $u_3(0, x) = 0$.

We have already checked that $\partial_t u_3 - \partial_x^2 u_3 = 0$. Therefore, u_3 is a solution to (1).

- e) To check that u_4 solves (1) we proceed similarly. The boundary conditions are satisfied, since $\sin(4\pi x) = 0$ at $x = 0, 1$. The initial condition is satisfied, since $e^{-16\pi^2 t} = 1$ at $t = 0$. Note also that,

$$\partial_t u_4 = -16\pi^2 u_4,$$

and,

$$\partial_x^2 u_4 = -16\pi^2 u_4.$$

Therefore, u_4 solves (1).

Problem 3. (Classification of PDEs)

Classify the following PDEs into linear/nonlinear, homogeneous/inhomogeneous:

- a) One dimensional heat equation,

$$\frac{\partial}{\partial t} u = \partial_x^2 u$$

- b) One dimensional heat equation on $[0, 1]$ with varying diffusion,

$$\frac{\partial}{\partial t} u = x(x-1) \partial_x^2 u$$

- c) The Laplace equation in polar coordinates ($u = u(r, \theta)$):

$$\frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r} \frac{\partial^2 u}{\partial \theta^2} = 0 .$$

- d) The Poisson equation describing torsion of prismatic bars:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + 3 = 0 .$$

- e) The convection-reaction equation, with a non-zero reaction exponent $n \in \mathbb{N}$:

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} + u^n = 0$$

- f) The bi-harmonic equation in two dimensions:

$$\frac{\partial^4 u}{\partial x^4} + 8 \frac{\partial^4 u}{\partial x^2 \partial y^2} + \frac{\partial^4 u}{\partial y^4} = 0 .$$

Solution.

Using the following definition: *A PDE is linear if it is of the first degree in the unknown function u and its partial derivatives. A linear PDE is homogeneous if its terms contain either u or one of its partial derivatives.*

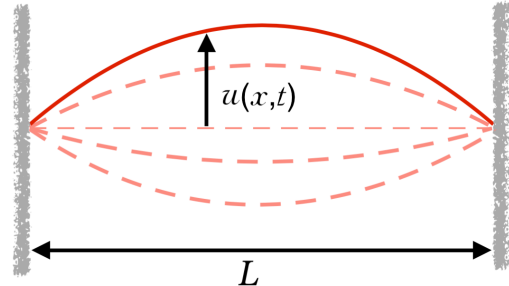
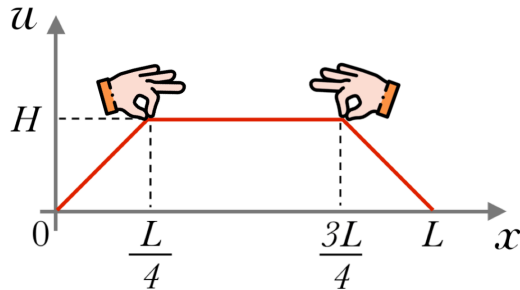
- a) This equation is **linear** and **homogeneous**.
- b) This equation is **linear** and **homogeneous**.
- c) The Laplace equation is **linear** and **homogeneous**.
- d) The Poisson equation is **linear** and **inhomogeneous**.
- e) The convection-reaction equation is **linear** if $n = 1$ and **nonlinear** if $n > 1$. It is also **homogeneous**.
- f) This 1-D heat equation is **linear** and **inhomogeneous**.
- g) The bi-harmonic equation is **linear** and **homogeneous**.

Problem 4. (Wave equation)

The small transversal displacements $u(x, t)$ of a stretched string satisfy the wave equation:

$$\frac{\partial^2 u}{\partial t^2} = \frac{FL}{m} \frac{\partial^2 u}{\partial x^2}, \quad (2)$$

where L and m are the string's length and mass, respectively, and F is the tension pulling the string (all three of them constant). With the string fixed on both sides, we get the boundary conditions $u(0, t) \equiv u(L, t) \equiv 0$. Let's say we grab the string at two points $x = L/4$ and $x = 3L/4$, and pull it up until reaching a height H . Starting from this trapezoidal shape at $t = 0$, the string is released and starts vibrating:



This initial condition for $u(x, t)$ can be written either in three separate linear parts, or also more compactly as

$$u(x, 0) = \left(\frac{1}{2} + \left| \frac{2x}{L} - 1 \right| - 1 \right) H. \quad (3)$$

Moreover, since the string is simply *released*, the velocity $\frac{\partial u}{\partial t}$ will be zero at $t = 0$.

- Determine the wave speed c in terms of the given constants F , L and m .
- Determine the vibration period in terms of F , m and L .
- Using separation of variables, find a general solution $u(x, t)$ for Eq. (2) fulfilling the boundary conditions.
- Finally, considering $L = 1$, apply the initial conditions to find the solution $u(x, t)$ of this initial-boundary value problem
Hint: if you want, you can use some symbolic integration tool to evaluate the necessary integrals.

Solution.

- Recalling the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2},$$

the wave speed is given by $c = \sqrt{\frac{FL}{m}}$.

- b) The period T is equal to the inverse of the frequency ν , which is defined for the fundamental mode as

$$\nu = \frac{c}{2L} = \frac{1}{2L} \sqrt{\frac{FL}{m}} = \frac{1}{2} \sqrt{\frac{F}{Lm}}.$$

Then the period is

$$T = 2\sqrt{\frac{Lm}{F}}.$$

- c) We can write $u(x, t)$ as $u(x, t) = F(x)G(t)$. Equation (2) becomes

$$F(x)G''(t) = c^2 F''(x)G(t),$$

which can be separated as

$$\frac{F''(x)}{F(x)} = \frac{G''(t)}{c^2 G(t)} = k$$

since x and t are independent of each other. We get two ordinary differential equations

$$F''(x) - kF(x) = 0 \tag{4}$$

$$G''(t) - c^2 kG(t) = 0, \tag{5}$$

The boundary conditions $u(0, t) = u(L, t) = 0$, i.e.

$$F(0)G(t) = 0 = F(L)G(t) = 0,$$

implies that $G \neq 0$. This is because $G(t) = 0$ is equivalent to $u(x, t) = 0$, which is of no interest. Then $F(0) = F(L) = 0$.

Solving Equation (4), we notice that $k = 0$ leads $F(x) = ax + b$. From the boundary conditions $b = 0$ and $a = 0$, and $F(x) = 0$, we have that, in the same way as $G(t) = 0$, this solution is of no interest. In the case $k > 0$, the solution is

$$F(x) = Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x},$$

from the boundary conditions, $A+B = 0$ and $Ae^{\sqrt{k}L} + Be^{-\sqrt{k}L} = 0$. Given $e^{\sqrt{k}L} - e^{-\sqrt{k}L} \neq 0$, the solution is $F(x) = 0$ due to $A = B = 0$.

Then k must be negative, and we can define $k = -p^2$. In this way, the solution of (4) is

$$F(x) = A \cos(px) + B \sin(px).$$

From the boundary conditions, we get that $A = 0$ and $\sin(pL) = 0$ (B can not be zero) with $p = \frac{n\pi}{L}$, $n \in \mathbb{N}$.

The solution for $F(x)$ is a family of solutions F_n defined as

$$F_n(x) = B \sin\left(\frac{n\pi}{L}x\right).$$

In particular we set $B = 1$.

We get a family of solutions G_n of (5)

$$G_n(t) = A_n \cos\left(c \frac{n\pi}{L}t\right) + B_n \sin\left(c \frac{n\pi}{L}t\right).$$

The general solution of Equation (2) is a family of solutions u_n given by

$$\begin{aligned} u_n(x, t) &= F_n(x)G_n(t), \\ u_n(x, t) &= \sin\left(\frac{n\pi}{L}x\right) \left(A_n \cos\left(c \frac{n\pi}{L}t\right) + B_n \sin\left(c \frac{n\pi}{L}t\right) \right). \end{aligned}$$

In general, this family of solutions does not satisfy the initial conditions. To ensure it does, we should use the Theorem of Superposition and consider a complete solution of the form

$$u(x, t) = \sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left(A_n \cos\left(c \frac{n\pi}{L}t\right) + B_n \sin\left(c \frac{n\pi}{L}t\right) \right).$$

d) Considering the initial condition (3), we have

$$\sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) A_n = H\left(\frac{1}{2} + \left|\frac{2x}{L} - 1\right| - 1 - \left|\frac{2x}{L} - 1\right| - \frac{1}{2}\right).$$

We can notice the coefficients $\{A_n\}_{n \geq 1}$ determine a Fourier series for the function

$$f(x) = H\left(\frac{1}{2} + \left|\frac{2x}{L} - 1\right| - 1 - \left|\frac{2x}{L} - 1\right| - \frac{1}{2}\right).$$

Then the coefficients $\{A_n\}_{n \geq 1}$ are computed by

$$A_n = 2 \int_0^1 H \left(\frac{1}{2} + ||2x - 1| - 1| - \left| |2x - 1| - \frac{1}{2} \right| \right) \sin(n\pi x) dx,$$

since $L = 1$. Using SymPy, the coefficients are

$$A_n = \frac{H}{n^2 \pi^2} (n\pi \cos(n\pi) - n\pi + 8 \sin(n\pi/4) - 8 \sin(n\pi/2) + 8 \sin(3n\pi/4) - 4 \sin(n\pi))$$

$$A_n = \frac{H}{n^2 \pi^2} (n\pi(-1)^n - n\pi - 8 \sin(n\pi/2))$$

$$A_{2s} = 0$$

$$A_{2s-1} = -\frac{H}{(2s-1)^2 \pi^2} ((2s-1)\pi + 4(-1)^{s+1})$$

From the first line to the second we have used $\cos(n\pi) = (-1)^n$, $\sin(n\pi) = 0$ and $\sin(\theta) = -\sin(\theta + \pi)$. The complete solution $u(x, t)$ satisfying the initial conditions is

$$u(x, t) = H \sum_{s=1}^{\infty} \sin((2s-1)\pi x) \left(\frac{((1-2s)\pi + 4(-1)^s) \cos(c(2s-1)\pi t)}{(2s-1)^2 \pi^2} \right).$$

The next exercise is optional and should not be handed in!

Problem 5. (Damped wave equation)

One way to include dissipation in wave modelling is via linear damping. In that case, the wave equation becomes

$$\frac{\partial^2 u}{\partial t^2} + 2\gamma \frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2},$$

in which c is the wave speed and $\gamma > 0$ is the damping parameter. As for the standard wave equation, we need boundary and initial conditions. Considering the following ones

$$u(0, t) = 0,$$

$$u(L, t) = 0,$$

$$u(x, 0) = f(x),$$

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0$$

and assuming $\gamma < c\pi/L$, use the separation of variables technique to show that the solution to this problem is

$$u(x, t) = \sum_{n=1}^{\infty} C_n e^{-\gamma t} \sin(\lambda_n x) \cos\left(t\sqrt{(c\lambda_n)^2 - \gamma^2}\right), \text{ with } \lambda_n := \frac{n\pi}{L}, n \in \mathbb{N},$$

and determine the constants C_n in terms of $f(x)$.

Solution.

Using separation of variables $u(x, t) = F(x)G(t)$, we get two ordinary differential equations

$$F''(x) + \lambda^2 F(x) = 0, \quad (6)$$

$$G''(t) + 2\gamma G'(t) + c^2 \lambda^2 G(t) = 0. \quad (7)$$

Considering the boundary conditions in the same way we did in Problem 3, the solution for (6) is given by a family of solutions of the form

$$F_n(x) = B \sin(\lambda_n x),$$

with $\lambda_n = \frac{n\pi}{L}$, $n \in \mathbb{N}$. For simplicity we can set $B = 1$.

To solve (7), we use its associated characteristic equation

$$\omega^2 + 2\gamma\omega + c^2 \lambda_n^2 = 0,$$

with $\omega_{1,2} = -\gamma \pm \sqrt{\gamma^2 - c^2 \lambda_n^2}$.

From the assumption, we have that $\gamma < c\pi/L$. Thus in general we have $\gamma < cn\pi/L$ for all $n \in \mathbb{N}$. Hence

$$\omega_{1,2} = -\gamma \pm i\sqrt{c^2 \lambda_n^2 - \gamma^2}.$$

Therefore the solution for (7) is a family of solutions

$$\begin{aligned} G_n(t) &= A_n e^{\omega_{1t}} + B_n e^{\omega_{2t}} = e^{-\gamma t} \left(A_n e^{i\sqrt{c^2 \lambda_n^2 - \gamma^2} t} + B_n e^{-i\sqrt{c^2 \lambda_n^2 - \gamma^2} t} \right) \\ G_n(t) &= e^{-\gamma t} \left(C_n \cos\left(\sqrt{c^2 \lambda_n^2 - \gamma^2} t\right) + D_n \sin\left(\sqrt{c^2 \lambda_n^2 - \gamma^2} t\right) \right). \end{aligned}$$

Considering the initial condition $\frac{\partial u}{\partial t}\big|_{t=0} = 0$, we get that $D_n = 0$, and

$$u_n(x, t) = e^{-\gamma t} \sin(\lambda_n x) \left(C_n \cos \left(\sqrt{c^2 \lambda_n^2 - \gamma^2} t \right) \right).$$

The coefficients C_n can be computed as

$$C_n = \frac{2}{L} \int_0^L f(x) \sin(\lambda_n x) dx.$$

The final solution is written as

$$u(x, t) = \sum_{n=1}^{\infty} C_n \sin(\lambda_n x) \cos \left(t \sqrt{(c \lambda_n)^2 - \gamma^2} \right) e^{-\gamma t}.$$